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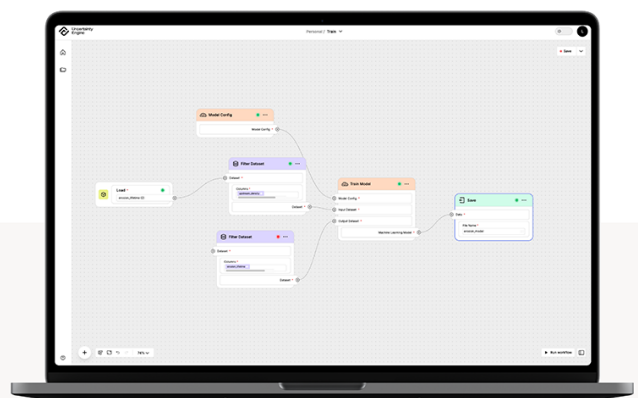


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Spatially dependent cluster dynamics model of He plasma surface interaction in tungsten for fusion relevant conditions

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Abstract

In fusion reactors, plasma facing components (PFC) and, in particular, the divertor will be irradiated with high fluxes of low-energy (~ 100 eV) helium and hydrogen ions. Tungsten is one of the leading candidate divertor materials for ITER and DEMO fusion reactors. However, the behaviour of tungsten under high dose, coupled helium/hydrogen exposure remains to be fully understood. The PFC response and performance changes are intimately related to microstructural changes, such as the formation of point defect clusters, helium and hydrogen bubbles or dislocation loops. Computational materials' modelling results are described here that investigate the mechanisms controlling microstructural evolution in tungsten. The aim of this study is to understand and predict sub-surface helium bubble growth under high flux helium ion implantation ($\sim 10^{22} \text{ m}^{-2} \text{ s}^{-1}$) at high temperatures (> 1000 K). We report results from a spatially dependent cluster dynamics model based on reaction–diffusion rate theory to describe the evolution of the microstructure under these conditions. The key input parameters to the model (diffusion coefficients, migration and binding energies, initial defect production) are determined from a combination of atomistic modelling and available experimental data. The results are in good agreement with results of an analytical model that is presented in a separate paper. In particular, it is found that the sub-surface evolution with respect to bubble size and concentration of the helium bubbles strongly depends on the flux and temperature.

Keywords: cluster dynamics, plasma facing materials, modelling, fusion, helium, tungsten

(Some figures may appear in colour only in the online journal)

1. Introduction

The divertor is a very important component in future fusion reactors with the function of extracting heat and helium ash from the plasma. Helium exposure with fluxes up to $10^{23} \text{ m}^{-2} \text{ s}^{-1}$ and wall temperatures between 1000 and 2000 K are expected in fusion reactors, including ITER and DEMO [1]. These severe operating conditions pose new material challenges. Among materials investigated for the divertor, tungsten is one of the most promising due to its high melting temperature and low erosion rate. Little understanding exists about the behaviour of helium in tungsten under these operating conditions. Several issues associated with helium exposure have been experimentally observed, such as blistering and flaking [2], changes in tritium retention [3], and the formation of tungsten nanofuzz [1].

In order to simulate the fate of tungsten under helium exposure at high doses and for large penetration depths, the use of continuum models is necessary. Cluster dynamics is one such model based on reaction rate theory combined with diffusion processes. It has been successfully applied to describe the evolution of the microstructure of different elements under various irradiation conditions in recent years [4–8]. The use of cluster dynamics modelling does have both physical as well as computational limitations; namely physical challenges due to the large number of physical quantities (diffusion coefficients and binding energies) to determine for each chemical species, and applied mathematics and computational challenges due to the large number of species that are modelled. In a previous article [9], we presented a cluster dynamics model to describe the microstructure evolution of tungsten under low-flux and low-energy helium

implantation. The model was benchmarked with several thermal helium desorption spectrometry (THDS) experiments and a good agreement was found between experimental and simulated results. Therefore, it appears to include a reasonable description of the properties and interactions of small mobile helium defects as well as small helium–vacancy clusters formed under low flux irradiation.

This article focuses on the extension of this model to the case of high flux helium irradiation typically encountered in fusion reactors. The novelty of the model is its ability to describe the evolution of the tungsten microstructure due to very high incoming fluxes of helium ions resulting in a rapid growth of large helium bubbles under the materials surface. We quickly introduce the model and present a method to deal with very large helium–vacancy clusters which can form during cluster growth under continuous helium implantation. We also address the issue of the validity of the dilute limit approximation at such doses. To provide further benchmarking to the model, we present results for two specific configurations of interest in tungsten irradiation under fusion relevant conditions: a tendril-like configuration, and a half space domain configuration. We also compare the results to an analytical model developed by Krashennnikov *et al* [10], and molecular dynamics (MD) simulations performed by Sefta *et al* [11].

2. Model description

2.1. Spatially dependent cluster dynamics model

We define the term species to include any unique combination of atoms or defects grouping to form a cluster in the material. The species considered in our model are interstitial helium atoms (denoted as He_1) and clusters of helium atoms (He_x , $x \geq 2$), single vacancies (V_1) and clusters or vacancies (V_y , $y \geq 2$, also called voids), single self-interstitial tungsten atoms (I_1) and clusters of self-interstitial atoms (SIA) (I_z , $z \geq 2$, also called interstitial loops), and helium–vacancy clusters ($\text{He}_x V_y$, $x \geq 1$, $y \geq 1$, also called gas bubbles when they reach a nanometre size. We do not consider helium–interstitial clusters or helium–vacancy–interstitial clusters in this work: although they appear feasible in iron [12], there is no experimental evidence to confirm the existence of these clusters in tungsten, and little simulation work has been performed to investigate the thermodynamics or kinetics of these possible clusters. Each individual species has specific physical properties (such as diffusion coefficient, binding energies and capture radius) and is therefore treated separately. Experiments and simulations show that helium–vacancy clusters can contain up to hundreds of vacancies and helium gas bubbles that can reach sizes in excess of tens of nanometres [13]. Therefore, the total number N of species to consider can easily reach $N > 1\,000\,000$.

Single mobile species can react to form clusters. Each cluster of helium atoms, vacancies and interstitials is considered as a separate species characterized by the type and number of helium atoms and point defects clustered. The model is then based on the mean field rate theory equations: the time and spatial evolution of the concentration $C_i(\vec{r}, t)$ is calculated for each species i . This evolution is dictated by four different factors: direct production through implantation,

diffusion in the tungsten matrix, reaction between species, and dissociation of species.

The evolution of the concentration of each species is described by a system of partial differential equations (PDEs) of the form

$$\begin{aligned} \frac{\partial C_i}{\partial t}(\vec{r}, t) = & D_i \nabla C_i(\vec{r}, t) \\ & + P_i(\vec{r}) \quad \text{for } i = 1, \dots, N \\ & + \text{Reaction}_i(C_1(\vec{r}, t), \dots, C_N(\vec{r}, t)) \\ & + \text{Dissociation}_i(C_1(\vec{r}, t), \dots, C_N(\vec{r}, t)) \end{aligned} \quad (1)$$

where $D_i \nabla C_i(\vec{r}, t)$ is the diffusion term due to a concentration gradient of the chemical species. D_i is the diffusion coefficient of the species considered. In this paper, our model assumes symmetry in directions perpendicular to the irradiation flux and therefore this term reduces to $D_i \frac{\partial^2 C_i}{\partial x^2}(x, t)$. Extension of the model to two or three dimensions to include the effect of grain boundaries or other spatial heterogeneities within the material is possible, but is not discussed further here.

$P_i(\vec{r})$ is the production term due to implantation (or irradiation). We use the program SRIM [14] which provides a reasonably accurate prediction of the implanted helium range distribution. For low-energy incident ions, no point defect is directly produced by irradiation. For high energy incident ions, SRIM can predict the depth and energy distribution of primary knock-on atoms produced, but cannot accurately predict instant clustering of defects due to immediate recombination. Therefore, we couple the PKA depth and energy distribution calculated from SRIM with MD simulation of displacement cascades at the relevant PKA energies to estimate the actual number of point defects and clusters of point defects created within the initial damage production when appropriate, and use equation (1) to model the subsequent diffusive evolution. Figure 1 shows the range probability density function obtained using SRIM for a helium flux with incoming energy $E_i = 100$ eV.

$\text{Reaction}_i(C_1(\vec{r}, t), \dots, C_N(\vec{r}, t))$ represents the change in concentration due to reactions between clusters of helium and point defects, and $\text{Dissociation}_i(C_1(\vec{r}, t), \dots, C_N(\vec{r}, t))$ represents the change in concentration due to dissociation of clusters into smaller ones. The reaction and dissociation terms are obtained from chemical reaction theory. For a reaction between two chemical species A and B :



$$\frac{\partial C_{AB}}{\partial t} = -\frac{\partial C_A}{\partial t} = -\frac{\partial C_B}{\partial t} = k_{A,B}^+ C_A C_B - k_{AB}^- C_{AB}. \quad (3)$$

The forward reaction rate constant can be calculated using the theory of diffusion-limited reactions [15]:

$$k_{A,B}^+ = 4\pi(r_A + r_B)(D_A + D_B) \quad (4)$$

r_A and r_B are the capture radius of species A and B and represent the distance at which the species react to form the product AB .

The backward dissociation rate constant can be calculated using chemical equilibrium principles [5]:

$$k_{AB}^- = k_{AB}^+ \frac{C_{Aeq} C_{Beq}}{C_{ABeq}} = \frac{1}{\Omega} k_{AB}^+ e^{-\frac{\Delta G^0}{k_b T}} = \frac{1}{\Omega} k_{AB}^+ e^{-\frac{E_b(A+B \rightarrow A+B)}{k_b T}}. \quad (5)$$

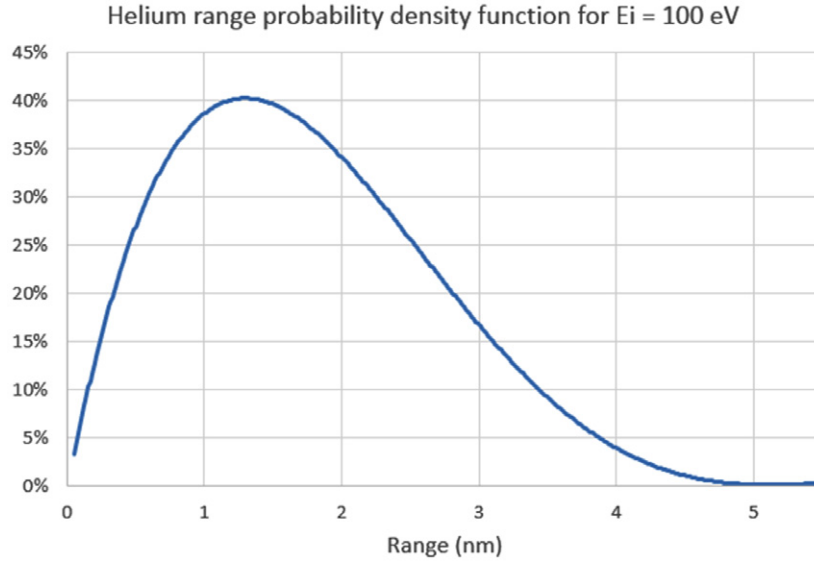


Figure 1. Range probability function of 100 eV helium in tungsten as calculated by SRIM [14].

Table 1. List of possible clustering reactions considered in this model.

Reaction	Formula	Comment
Helium clustering	$\text{He}_x + \text{He}_y \rightarrow \text{He}_{x+y}$	Only possible for small clusters ($x + y < 10$) due to trap mutation
Vacancy clustering	$V_x + V_y \rightarrow V_{x+y}$	Di-vacancy stable with MD data, not stable with DFT data
Interstitial clustering	$I_x + I_y \rightarrow I_{x+y}$	Interstitial concentration remains negligible for low energy irradiation
Interstitial–vacancy annihilation	$I_x + V_y \rightarrow \begin{cases} I_{x-y} & \text{if } x > y \\ V_{y-x} & \text{if } x < y \\ \emptyset & \text{if } x = y \end{cases}$	
Helium–vacancy clustering	$\text{He}_x + V_y \rightarrow \text{He}_x V_y$	
He/V cluster growth through helium absorption	$\text{He}_x V_y + \text{He}_z \rightarrow \text{He}_{x+z} V_y$	
He/V cluster growth through single vacancy absorption	$\text{He}_x V_y + V \rightarrow \text{He}_x V_{y+1}$	
He/V cluster reduction through interstitial absorption	$\text{He}_x V_y + I_z \rightarrow \text{He}_x V_{y-z}$	

Ω is the atomic volume, k_b is the Boltzmann constant, T is the temperature and $E_b(AB \rightarrow A + B)$ is the binding energy of the reaction $AB \rightarrow A + B$. The description of the clustering and dissociation reactions considered in this work are provided in tables 1 and 2, respectively.

2.2. Assumptions

In the above description of the cluster dynamics model, we made three major assumptions:

- *Diffusion controlled reactions.* Two species are assumed to react instantaneously when they are at a certain distance of each other. This is referred to as diffusion-limited reactions, since the rate of transport of the reactants through the tungsten matrix is the limiting step in the reaction and not the rate of formation of the product. It is observed in MD simulations that reactants indeed instantaneously form a product when they come in close contact of each other [11, 16].
- *Dilute limit approximation.* In the derivation of the capture constant given by equation (4), it is assumed

that the two reacting species are randomly and uniformly distributed, and can freely diffuse before reacting with each other. This implies that the model does not take into account spatial correlations between species. Hence, the model is only valid in the dilute limit: the volume occupied by the species must remain small compared to the volume of the surrounding tungsten matrix. If $V_f = \sum_{i=1}^N 4/3\pi r_i^3 C_i$ denotes the total volume fraction of the species, we need $V_f \ll 1$ at all times. This can become a concern under fusion relevant conditions at very high fluxes ($\sim 10^{22} \text{ m}^{-2} \text{ s}^{-1}$) and fluences ($\sim 10^{25} \text{ m}^{-2}$) and we shall therefore monitor this parameter at all times.

- *Dimensionality of motion.* The derivation of the capture constant also assumes that the species can freely diffuse in three dimensions. MD simulations show that this is indeed the case for helium interstitials as well as vacancy and SIA defects. Figure 2 shows an example of this three-dimensional motion for a single SIA at 1000 K. Due to the frequent rotations of the SIA, the motion can be

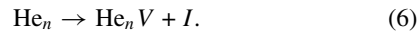
Table 2. List of possible dissociation reactions considered in this model.

Dissociation reaction	Formula	Comment
Helium dissociation	$\text{He}_x \text{V}_y \rightarrow \text{He}_{x-1} \text{V}_y + \text{He}$	Single He dissociates from cluster
Vacancy dissociation	$\text{He}_x \text{V}_y \rightarrow \text{He}_x \text{V}_{y-1} + \text{V}$	Single vacancy dissociates from cluster
Trap mutation	$\text{He}_x \text{V}_y \rightarrow \text{He}_x \text{V}_{y+1} + \text{I}$	High pressure He/V cluster eject a lattice atom to create a single interstitial

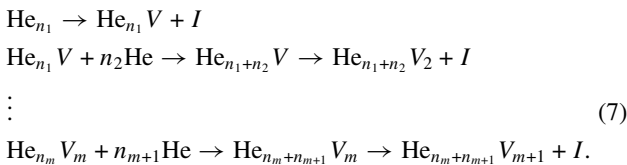
considered three-dimensional at a larger time scale. While it is true that SIA clusters rotate much less frequently, it is not an issue in this work since the low energy of the incident helium ions considered here do not allow direct production of large interstitial clusters, nor do we observe significant concentrations of SIA clusters formed in this model. Indeed, most SIA created through trap mutation annihilate with existing vacancies or diffuse rapidly to annihilate with the free surface which acts as a sink.

2.3. Trap mutation phenomenon

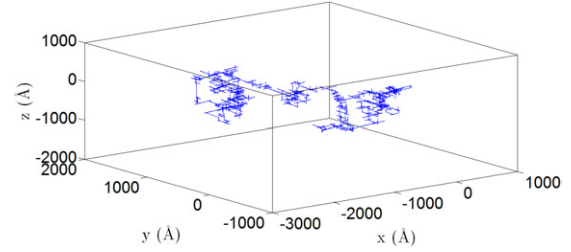
At high fluxes and temperatures, the following growth mechanism is proposed based on observations from MD simulations [11, 16]: injected helium interstitials diffuse in the tungsten matrix and react with each other, thus forming small helium–interstitial clusters. These small interstitial clusters grow by self-trapping of additional helium interstitials and helium–interstitial clusters, until they reach a critical, temperature dependent size at which a trap mutation phenomenon occurs. The growing, over-pressurized helium–interstitial cluster ejects a surrounding tungsten atom to form a helium–vacancy cluster, in a process we define as trap mutation:



The tungsten self-interstitial atom formed is free to diffuse towards the free surface or react with other species, while the immobile $\text{He}_n \text{V}$ cluster formed can keep growing by absorption of more mobile helium atoms. When it reaches a sufficient size, trap mutation can occur again, and the growth mechanism can be described as:



Trap mutation events ultimately determine which species are considered in the model, since only a limited number of helium can be packed together in a given sized helium–vacancy cluster. To determine this limit, we are using the results of molecular statics simulations by Juslin *et al* [17]. At 0 K, the results of the minimizing the energy for a given helium–vacancy cluster can lead to a static trap mutation event. When this happens, we consider that species to be the limiting number of helium for that amount of vacancy. Any reaction with other mobile helium species will lead to a trap mutation event. Figure 3 shows the limiting number of helium per vacancy at which trap mutation occurs, as computed using molecular statics simulations. We see that past a certain size (~ 10 vacancies), the limiting

**Figure 2.** 100 ns trajectory of a single self-interstitial atom from MD simulations at 1000 K.**Figure 3.** Limiting number of helium per vacancy at which trap mutation occurs at 0 K.

number of helium per vacancy at which trap mutation occurs stabilizes to a value of approximately 4. Therefore, under fusion relevant plasma surface interaction conditions, very large helium–vacancy clusters can quickly form in the tungsten matrix even in the absence of pre-existing traps for the mobile helium.

2.4. Extension of the model to large clusters

Since the number of helium and vacancies per species can quickly grow to thousands, it becomes very challenging to use the model equations for each different species: first, we need to determine the binding energies for helium dissociation, vacancy dissociation and trap mutation for every single species, and second we need to solve the model equations for a very large number of species ($N > 1\,000\,000$).

The first problem can be solved by extrapolating the binding energies of large helium–vacancy clusters based on the binding energies computed using MD. The second problem can be alleviated in a similar way: since very large over-pressurized helium–vacancy clusters have a similar behaviour, we are more interested in general quantities such as total cluster concentration, average size and number of helium or volume fraction occupied by helium–vacancy clusters rather than in

the detailed distribution of the concentration of each helium–vacancy cluster.

To explain this in detail, let us order the species considered from 1 to N . We have computed the binding energies using MD for each species up to size N , and we are able to solve the model equations for each of these cluster species. The largest species explicitly considered corresponds to a helium–vacancy cluster with 100 vacancies and 500 helium. The cluster dynamics equations can be written

$$\begin{aligned} \frac{\partial C_1}{\partial t} &= D_1 \frac{\partial^2 C_1}{\partial x^2} + P - 2k_{1,1}^+ C_1^2 - \sum_{i=2}^N k_{1,i}^+ C_1 C_i \\ &\quad - \sum_{i=N+1}^{\infty} k_{1,i}^+ C_1 C_i \\ \frac{\partial C_2}{\partial t} &= D_2 \frac{\partial^2 C_2}{\partial x^2} - k_{1,2}^+ C_1 C_2 + k_{1,1}^+ C_1^2 \\ &\vdots \\ \frac{\partial C_N}{\partial t} &= -k_{1,N}^+ C_1 C_N + k_{1,N-1}^+ C_1 C_{N-1} \\ \frac{\partial C_{N+1}}{\partial t} &= -k_{1,N+1}^+ C_1 C_{N+1} + k_{1,N}^+ C_1 C_N \\ &\vdots \end{aligned} \quad (8)$$

Now let us define the total bubble concentration and average bubble helium content, radius and capture constant:

$$C_b = \sum_{i=N+1}^{\infty} C_i \quad (9)$$

$$\langle i_b \rangle = \frac{\sum_{i=N+1}^{\infty} i C_i}{C_b} \quad (10)$$

$$\langle r_b \rangle = \frac{\sum_{i=N+1}^{\infty} r_i C_i}{C_b} \approx f(\langle i_b \rangle) \quad (11)$$

$$\langle k_b^+ \rangle = \frac{\sum_{i=N+1}^{\infty} k_{1,i}^+ C_i}{C_b} = 4\pi D_1 (r_1 + \langle r_b \rangle). \quad (12)$$

The evolution equations become

$$\frac{\partial C_1}{\partial t} = D_1 \frac{\partial^2 C_1}{\partial x^2} + P - 2k_{1,1}^+ C_1^2 - \sum_{i=2}^N k_{1,i}^+ C_1 C_i - \langle k_b^+ \rangle C_1 C_b \quad (13)$$

$$\frac{\partial C_2}{\partial t} = D_2 \frac{\partial^2 C_2}{\partial x^2} - k_{1,2}^+ C_1 C_2 + k_{1,1}^+ C_1^2 \quad (14)$$

$$\vdots \quad (15)$$

$$\frac{\partial C_n}{\partial t} = -k_{1,n}^+ C_1 C_n + k_{1,n-1}^+ C_1 C_{n-1} \quad (16)$$

$$\frac{\partial C_b}{\partial t} = k_{1,N}^+ C_1 C_N \quad (17)$$

$$\frac{\partial (\langle i_b \rangle C_b)}{\partial t} = (N+1) k_{1,N}^+ C_1 C_N + \langle k_b^+ \rangle C_1 C_b. \quad (18)$$

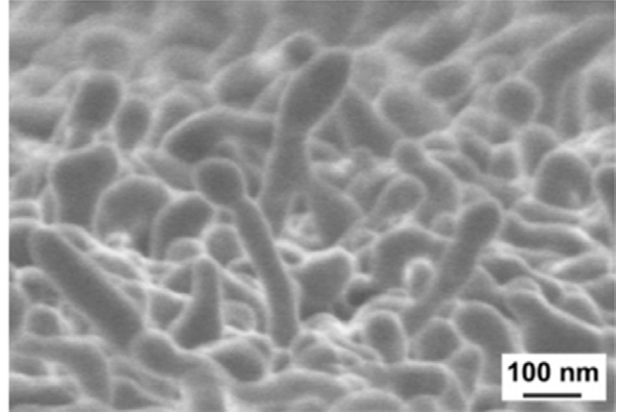


Figure 4. SEM micrograph showing a tungsten surface after 50 eV helium ion irradiation up to a fluence of $1.1 \times 10^{25} \text{ m}^{-2}$ at 1400 K. Reproduced with permission from [1]. Copyright 2009 IAEA.

This approach to extending the model description to a larger He–vacancy cluster phase space consists of two main approximations:

- First, we assume

$$\begin{aligned} \langle r_b \rangle &= r(\text{He}_{\langle i_b \rangle} V_{\langle i_b \rangle/4}) = r_{\text{He}_0 V_1} + \left(\frac{3}{4\pi} \frac{a_0^3}{2} \frac{\langle i_b \rangle}{4} \right)^{1/3} \\ &\quad - \left(\frac{3}{4\pi} \frac{a_0^3}{2} \right)^{1/3} \end{aligned} \quad (19)$$

where a_0 is the lattice parameter ($a_0 = 0.318 \text{ nm}$ at $T = 1000 \text{ K}$). This equation means that we assume that the average radius of the bubbles is the radius of a helium–vacancy cluster with the average helium content $\langle i_b \rangle$. Since figure 3 shows we have approximately one trap mutation event, and thus one additional vacancy, for every 4 helium absorbed for large helium–vacancy clusters, that radius is then computed using the classical radius formula for a vacancy cluster containing $\langle i_b \rangle/4$ vacancies. Due to this approximation, the value of $\langle k_b^+ \rangle$ is not exact anymore and so the conservation of helium atoms is altered. This alteration depends on the actual distribution of large helium–vacancy clusters in the phase space: the more the distribution is skewed towards smaller and larger clusters than the average, the more questionable becomes this approximation. However, numerical experiments comparing the detailed model with an explicit account of large helium–vacancy clusters and the extended model show that the actual distribution of helium–vacancy clusters under constant irradiation conditions has a narrow Gaussian shape centred on the average value $\langle C_b \rangle$. Therefore, we expect relatively little impact of the extended model on the conservation of helium atoms under these conditions.

- The second approximation is that we neglect dissociation of large helium–vacancy clusters ($\geq \text{He}_{250} V_{50}$ in this paper). The binding energy for trap mutation events is indeed lower than the binding energy for helium or vacancy dissociation events for large over-pressurized helium–vacancy clusters. Therefore the frequency of helium and vacancy dissociation events is negligible

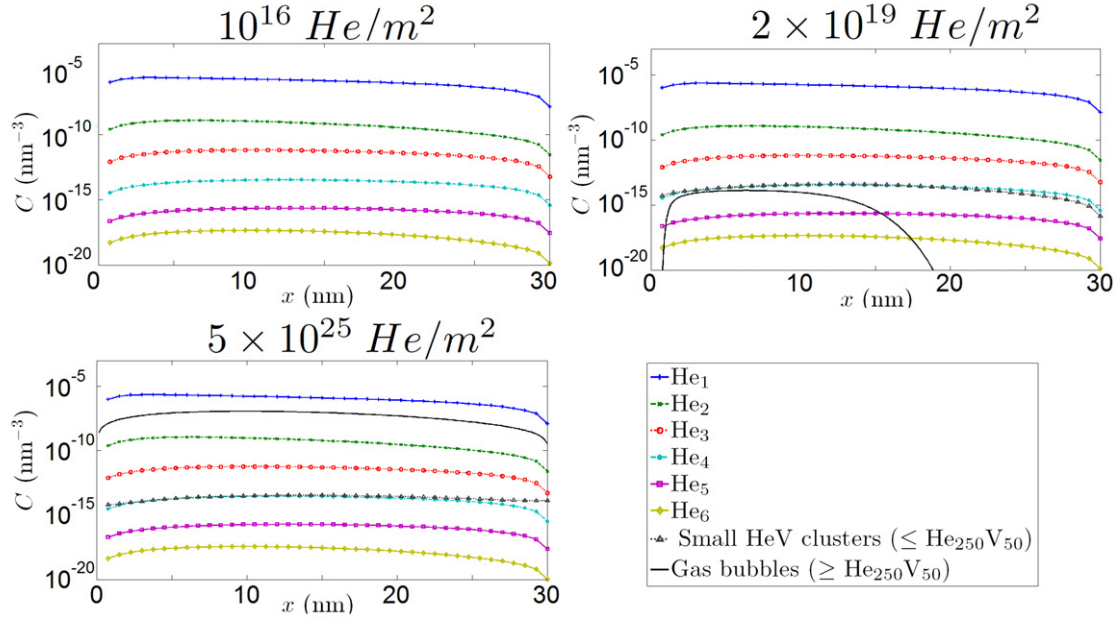


Figure 5. Concentration of mobile helium clusters and immobile helium–vacancy clusters in a tendril after $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium irradiation at 1000 K up at three different fluences.

compared to the frequency of trap mutation events, regardless of the temperature range.

2.5. Solver

The number N of species considered, and hence of cluster dynamics equations, can grow very large ($N > 10\,000$) as we need to consider larger and larger helium–vacancy clusters due to bubble growth. We developed a parallel solver to efficiently deal with the computational challenge that is solving the large system of coupled non-linear PDEs. The simple 1D geometry makes a good case for the use of finite difference methods, and we use an implicit Runge–Kutta scheme due to the stiffness of the diffusion and reaction operators. The key to efficiently solve the resulting non-linear equations is to notice that the vast majority of the species considered are immobile and hence their cluster dynamics evolution equation is an ordinary differential equation (ODE). These ODEs are local: the reaction term at a given grid point only depends on the other species concentration at the same grid point. Therefore, we can devise an algorithm to partition the system into a reduced system of equations with only the few mobile species, and an independent large system of equations local to the grid. The first system is typically very small and does not grow with an increasing number of helium–vacancy clusters considered since the number of mobile species is fixed, so it is easily solved. The second system is very large and can be solved using a parallel implementation which solves the system independently on each point of the grid, with therefore excellent parallel speedup and memory requirements properties. To accurately describe the evolution of the materials microstructure, we use a relative tolerance of 10^{-3} in the solver. The complete description of the solver is outside of the scope of this paper and is the object of a separate paper in preparation [18].

3. Results

3.1. Tendril geometry

Several experimental studies show a severe change in the surface morphology of tungsten under fusion relevant conditions of temperature $\geq 1000 \text{ K}$, helium flux $\geq 10^{22} \text{ m}^{-2} \text{ s}^{-1}$, fluence $\geq 10^{25} \text{ m}^{-2}$ and incident helium energy $\leq 100 \text{ eV}$ [1, 19]. At temperature above 1000 K, a nanometre-sized fibre structure, or tungsten ‘nanofuzz’, is formed consisting of tungsten tendrils filled with nanometre-sized gas bubbles separated by holes in the tungsten surface. Figure 4 shows an example of this nanofuzz structure, as reproduced from [1].

We have applied our continuum reaction–diffusion model to simulate helium irradiation in a tendril under different implantation conditions and study the impact of important parameters such as the irradiation temperature or flux. The tendril geometry is assumed to have a simple rectangular shape, with a length much greater than its width. Assuming axial symmetry, our one-dimensional code thus models helium implantation into the direction perpendicular to a side of the tendril.

3.1.1. Standard case. We present results for a standard case with the following parameters: uniform irradiation on a tendril side with flux $= 10^{22} \text{ m}^{-2} \text{ s}^{-1}$, fluence up to $5 \times 10^{25} \text{ m}^{-2} \text{ s}^{-1}$, temperature $= 1000 \text{ K}$, tendril width $\Delta = 30 \text{ nm}$ and length $L = 100 \mu\text{m}$. In such a tendril, the volume, V , is $9 \times 10^7 \text{ nm}^3$ and the presence of a single species corresponds to a concentration of $C \sim 10^{-8} \text{ nm}^{-3}$.

Figure 5 presents the results of the cluster dynamics code for this standard case. Each plot describes the concentration of the different types of helium–vacancy clusters as a function of the distance to the surface in the tendril (the surface lies at $x = 0$ and $x = 30 \text{ nm}$). The evolution from a plot to the next one corresponds to an increase in the implanted

helium amount, from $10^{16} \text{ m}^{-2} \text{ s}^{-1}$ to the maximum of $5 \times 10^{25} \text{ m}^{-2} \text{ s}^{-1}$.

We observe on the lowest fluence plot that there is a rapid clustering of implanted single interstitial helium atoms into small mobile helium clusters (He_1 to He_6) which diffuse further in the tendril. These mobile species quickly reach a steady-state value resulting from a balance between diffusion towards the free surface at $x = 0$ and $x = 30 \text{ nm}$, compared with reactions to form larger clusters.

The two larger fluence plots show that as these clusters react to form larger helium–vacancy clusters through the trap mutation process discussed in section 2.3, we see an increase in concentration of relatively small ($\leq \text{He}_{250} \text{V}_{50}$) helium–vacancy clusters. These clusters serve as the nuclei that grow into larger gas bubbles. The continuous flux of incident helium atoms causes both an increase in the concentration of gas bubble as well as the growth of these bubbles. At the end of the simulation (at a fluence of $5 \times 10^{25} \text{ m}^{-2}$), the concentration of gas bubbles remains relatively small ($C_{\text{bubbles}} \approx 10^{-8} \text{ nm}^{-3}$, which corresponds to one bubble in the whole tendril) and the concentration of small mobile helium clusters is negligible. This means that the tendril contains at most one gas bubble and that the vast majority of the helium is present only as single interstitial helium atoms.

3.1.2. Influence of temperature. Figure 6 shows the effect of varying the temperature. Each plot again depicts the concentration of the different types of helium–vacancy clusters as a function of the distance to the surface in the tendril (the surface lies at $x = 0$ and $x = 30 \text{ nm}$) at the maximum fluence of $5 \times 10^{25} \text{ m}^{-2} \text{ s}^{-1}$.

The bottom plot presents results for a simulation at a temperature of 1500 K. At this temperature, the mobility of helium in tungsten is greatly enhanced. This increases the diffusion and loss of helium at the surface, and at the same time increases the clustering rate of helium atoms. These results show that the diffusion effect dominates, as a much smaller fraction of implanted helium is available for clustering and hence we observe a negligible concentration of helium bubbles in the tendril compared to the results at 1000 K.

The top plot shows the opposite effect for a simulation at a temperature of 500 K. The mobility of helium decreases so that the losses due to diffusion towards the free surface are greatly reduced and much more helium is available to form helium–vacancy clusters, even though the clustering rate is lower. We observe that large gas bubbles are predicted to form under these irradiation conditions. In fact, most of the helium present in the tendril is contained in gas bubbles at this fluence and temperature.

3.1.3. Influence of helium flux. Figure 7 shows the effect of varying the incoming helium flux for a given irradiation temperature of 1000 K. We observe that decreasing temperature has the same effect as increasing the incoming helium flux; namely, in the first case, the mobility of the helium clusters decreases so that their concentration increases, and cluster growth is enhanced; while in the latter case, the concentration of single helium is increased which in turn increases the reaction rate with mobile helium clusters, again resulting in enhanced growth of large helium–vacancy clusters through trap mutation.

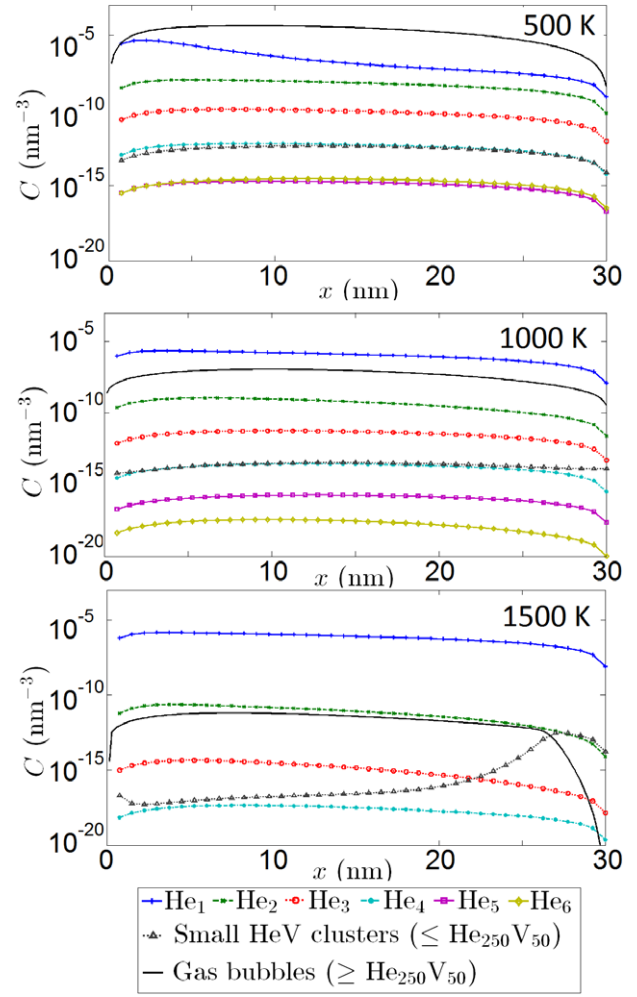


Figure 6. Concentration of mobile helium clusters and immobile helium–vacancy clusters in a tendril after $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium irradiation at a fluence of $5 \times 10^{25} \text{ m}^{-2}$ at three different temperatures: 500 K (top), 1000 K (middle), 1500 K (bottom).

3.1.4. Comparison with an analytical model. We can compare these results with an analytical helium cluster/bubbles dynamics model developed by Krashenninikov *et al* [10]. The analytical model assumption of constant uniform concentration of single mobile helium atoms and steady-state concentrations of mobile helium atom clusters in the tendril geometry is justified by our model results. The values of these steady-state concentrations, as shown in figure 8, are in excellent agreement with the values given by our model in figure 5. Figure 9 also compares the time required to produce a single immobile cluster as a function of the helium flux in both models at 1000 K. Again, an excellent agreement is found between the analytical model of Krashenninikov and our simulation results. This tends to confirm that large gas bubbles are not predicted to form in existing tendril geometries for temperatures above 1000 K and fluxes below $10^{22} \text{ m}^{-2} \text{ s}^{-1}$.

3.1.5. Summary and discussion. The results presented in this section indicate that the conditions to form helium gas bubbles in an originally defect-free tendril are a temperature lower than 1000 K and/or a flux higher than $10^{22} \text{ m}^{-2} \text{ s}^{-1}$.

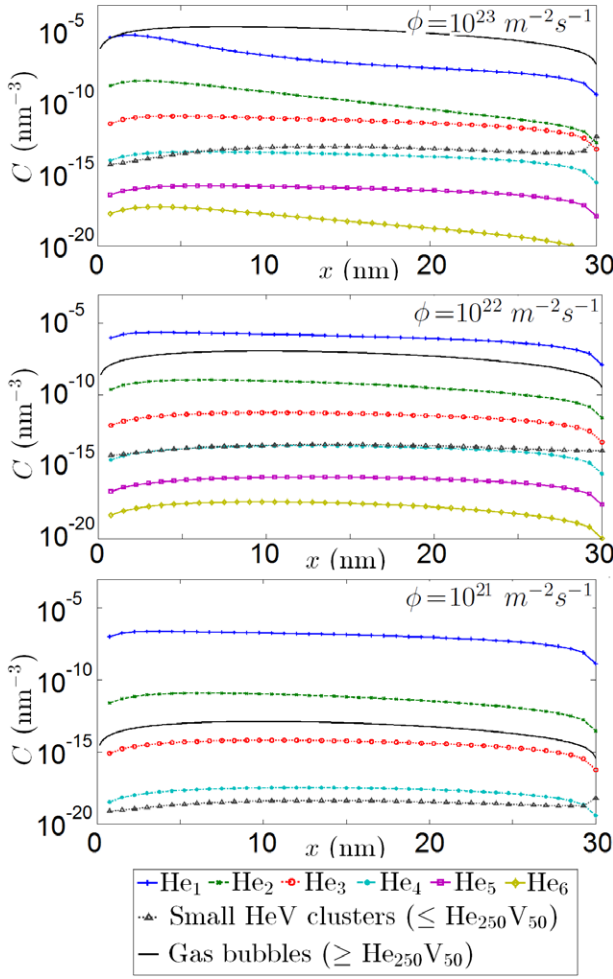


Figure 7. Concentration of mobile helium clusters and immobile helium–vacancy clusters in a tendril after 100 eV helium irradiation at 1000 K at a fluence of $5 \times 10^{25} \text{ m}^{-2}$ at three different fluxes: $10^{23} \text{ m}^{-2} \text{ s}^{-1}$ (top), $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ (middle), $10^{21} \text{ m}^{-2} \text{ s}^{-1}$ (bottom).

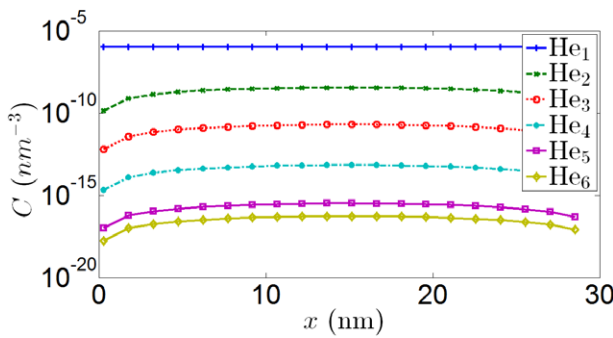


Figure 8. Steady-state concentrations of mobile helium clusters in a tendril for $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium implantation at 1000 K. Reproduced with permission from [10]. Copyright 2014 IAEA.

Since experiments show a high concentration of nanometre-sized bubbles in the tendrils which are formed at temperatures above 1000 K [20] and fluxes in the range of $10^{22} \text{ m}^{-2} \text{ s}^{-1}$, our results appear to indicate that defects are formed in the material before the tendrils are formed. In other words, we show that the microstructure of the tungsten material is strongly modified

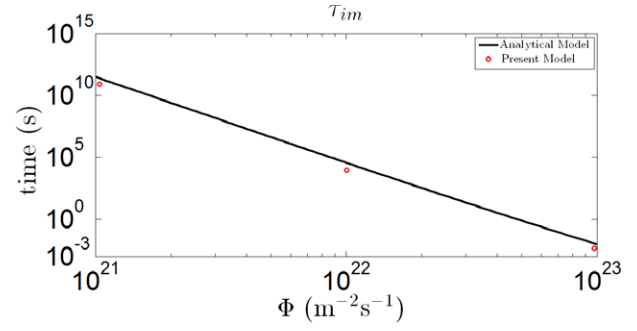


Figure 9. Time required to produce a single immobile helium cluster in the tendril as a function of helium implantation flux at 1000 K for both models.

before surface morphology changes such as tendril formation appear. To be able to describe nanofuzz formation, we should then first focus on describing the microstructure evolution in an irradiated tungsten material with a perfect surface, which is the topic of the next section.

3.1.6. Model limitations. The cluster dynamics model indicates that helium–vacancy defects are formed deeper in the material before surface morphology changes occur. However, we would like to discuss the current model limitations when applied to the tendril geometry: first, since our model is one dimensional and assumes symmetry in the other directions, the effect of free surfaces in a tendril geometry is underestimated. This would indicate an even smaller growth rate of large gas bubbles inside the tendrils. Second, we cannot accurately model the tendril evolution once large gas bubbles are formed: MD simulations indicate that large over-pressurized gas bubbles with a radius of several nanometres can burst when close to the surface [11], as is the case for the 30 nm wide tendril, and induce significant morphology changes. Third, we do not take into account pre-existing impurities in the tungsten which could act as nucleation sites for helium–vacancy clusters before self-trapping occurs, and hence influence both bubble nucleation and growth. These shortcomings are being addressed by a new three-dimensional cluster dynamics code being developed as part of a SciDAC project on plasma surface interactions [21], which should be able to provide much better predictions for nanofuzz or tendril structures. The present code is thus more adapted to predict the results of initial helium irradiation in the tungsten material before significant morphology changes appear, as we see in the next section.

3.2. Half-slab geometry

In this section, we provide model results for the case of helium implantation into a perfect tungsten surface. The geometry is that of an infinite half-slab, and the species are allowed to diffuse deep into the material, with a free surface at $x = 0$.

3.2.1. Standard case. Figure 10 shows results for an implantation flux of $\phi = 10^{22} \text{ m}^{-2} \text{ s}^{-1}$ and a temperature of 1000 K. We observe that helium-interstitial atoms diffuse deep into the tungsten material (several micrometres) within relatively short times. Past the implantation layer of a few

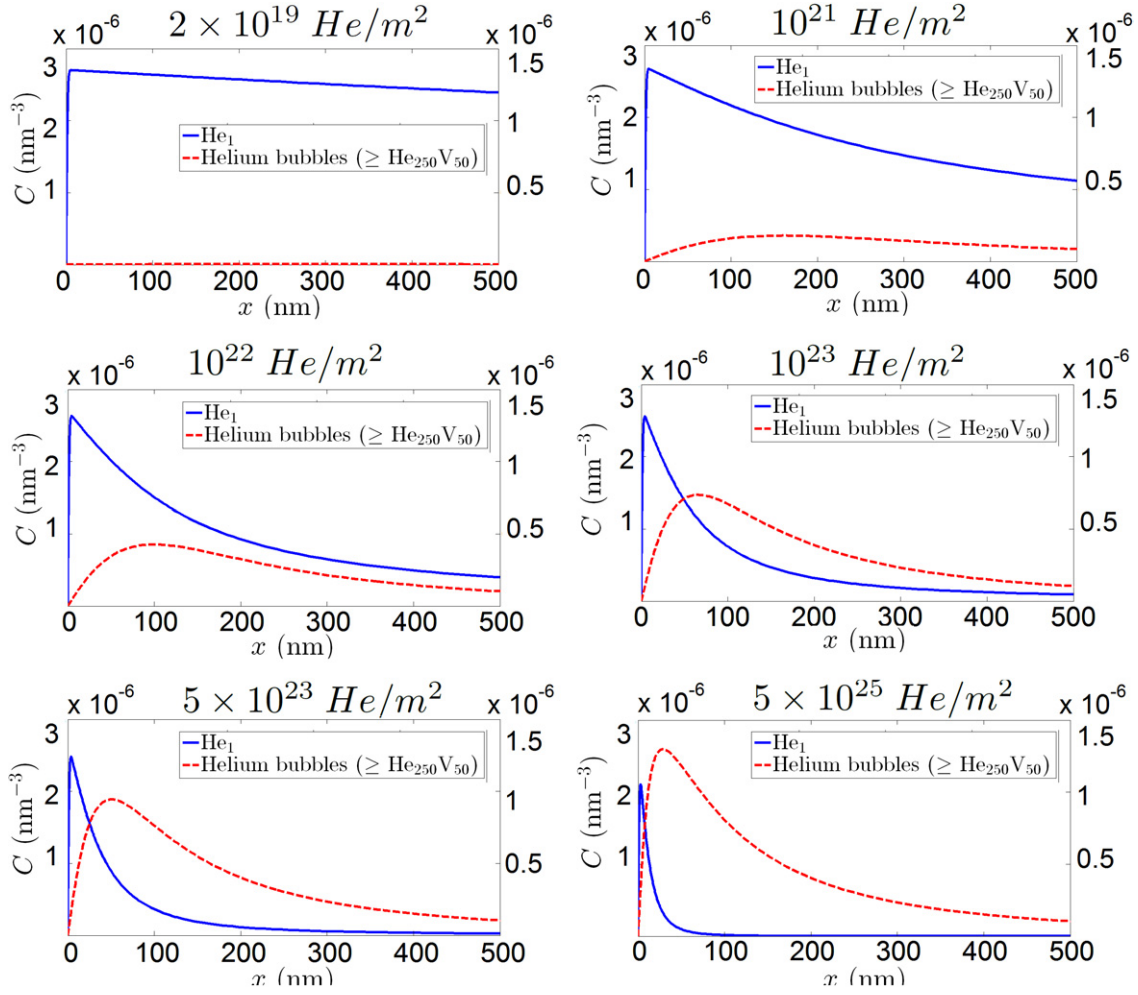


Figure 10. Concentration of helium-interstitial atoms and helium bubbles after $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium irradiation at 1000 K up to a fluence of $5 \times 10^{25} \text{ m}^{-2}$.

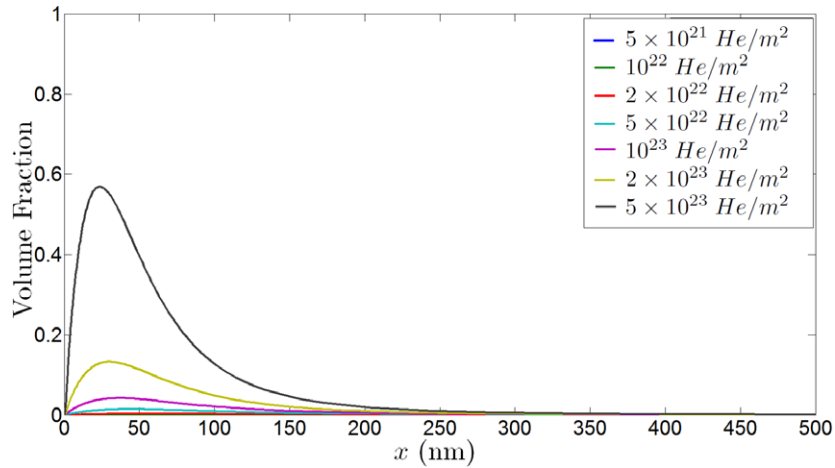


Figure 11. Volume fraction of helium species under the tungsten surface after $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium implantation at 1000 K up to a fluence of $5 \times 10^{23} \text{ m}^{-2}$.

nanometres, the helium-interstitial concentration is almost constant as a result of very fast helium diffusion in tungsten. As the fluence increases, self-trapping of helium-interstitial atoms create mobile helium clusters of increasing size. These clusters continue to grow by trapping more helium atoms and

become larger, immobile helium–vacancy clusters (He_xV_y) through the trap mutation process. The peak concentration of these helium bubbles is initially located far below the surface ($\sim 200 \text{ nm}$). As more helium interstitials are implanted and continue to interact and self-trap, as well as diffuse to and

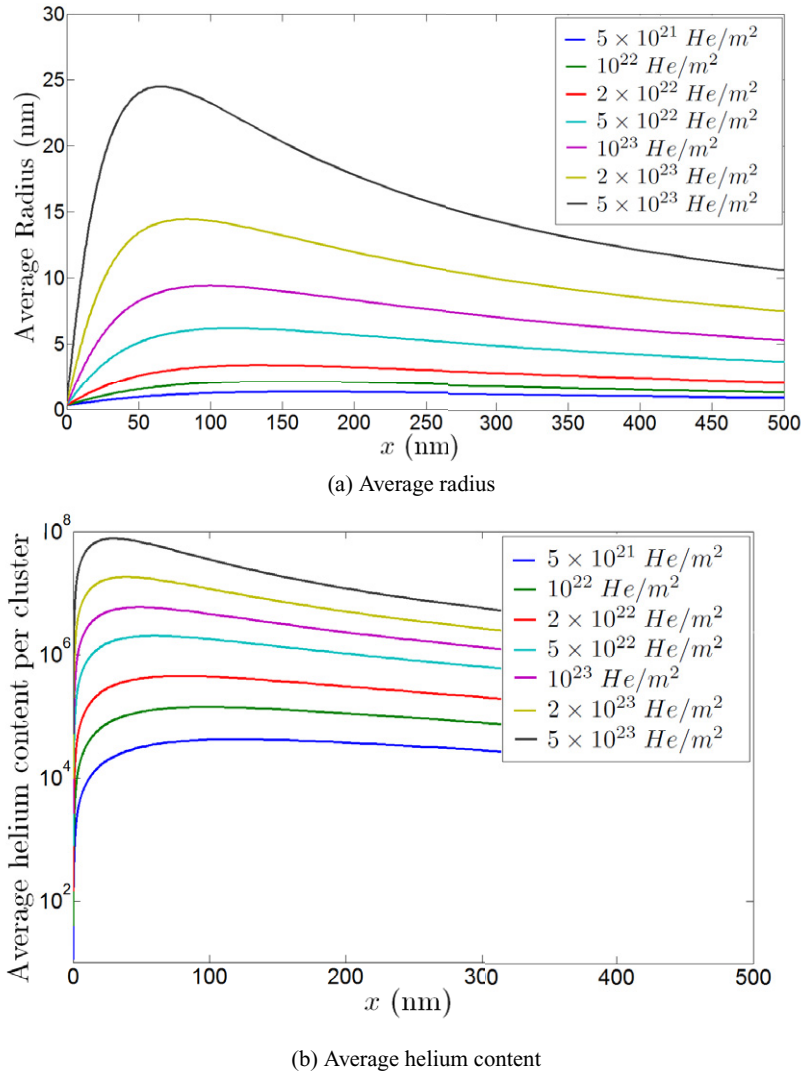


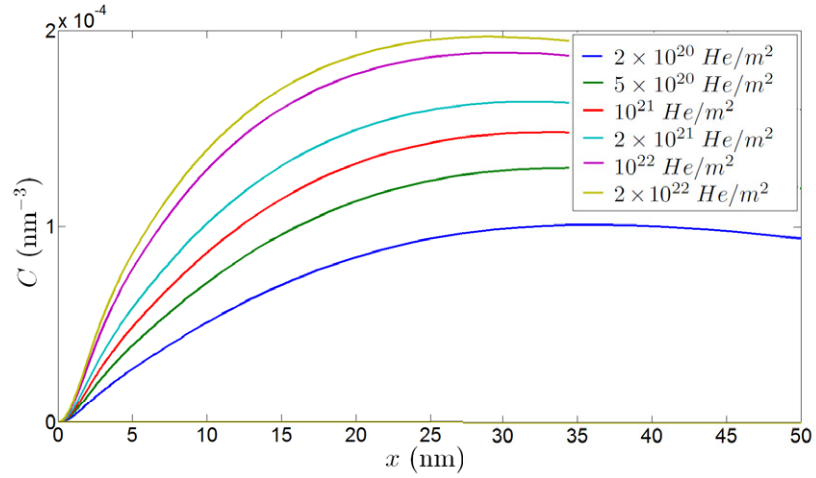
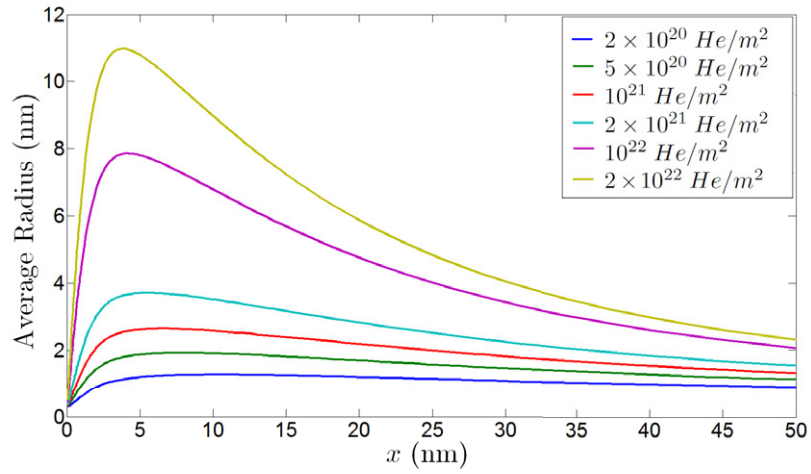
Figure 12. Average radius (a) and helium content (b) under the tungsten surface after $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium implantation at 1000 K up to a fluence of $5 \times 10^{23} \text{ m}^{-2}$.

react with the existing He_xV_y clusters, the helium bubbles concentration increases to reach a peak of $1.5 \times 10^{-6} \text{ nm}^{-3}$ at an implanted fluence of $5 \times 10^{25} \text{ m}^{-2}$. We also observe that these helium bubbles act as a ‘plug’ preventing the deeper diffusion of helium interstitials into the tungsten material: this plug of large helium bubbles slowly moves towards the surface as incoming helium interstitials react at the front of the plug, while we observe a depletion of the helium content in the material beyond the front of the plug. At the end of the simulation, the plug is located 50 nm below the surface, while it was initially formed several hundred nanometres below.

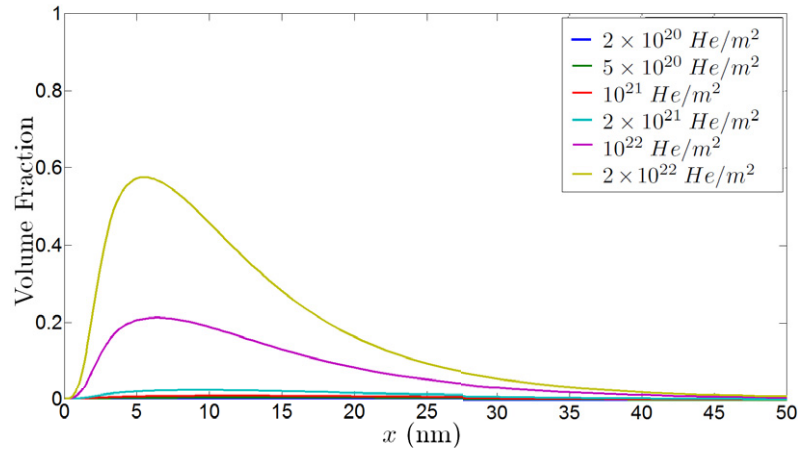
At the largest implanted fluence, the over-pressurized gas bubbles will eventually be close enough to the surface and reach a size which will produce bubble bursting and hence cause significant surface morphology changes, as observed in MD simulations [11]. However, this model does not yet include such bubble bursting events. Moreover, figure 11 shows another concern for the cluster dynamics model: even before bubble bursting and the associated morphology changes at the surface become a concern, the volume fraction occupied by helium species in the tungsten material becomes very large.

Hence, we can expect that the dilute limit approximation which ensures the validity of the cluster dynamics model breaks down due to the strong local interactions between species. In this regime, phenomena such as bubble coalescence are expected to become dominant. For a flux of $\phi = 10^{22} \text{ m}^{-2} \text{ s}^{-1}$ and a temperature of 1000 K, the dilute limit approximation breaks down at a fluence of approximately $5 \times 10^{23} \text{ m}^{-2}$. Figure 12 shows the corresponding growth of helium–vacancy clusters in the same simulation. At a fluence of $5 \times 10^{23} \text{ m}^{-2}$, 25 nm wide helium bubbles have formed containing millions of helium atoms at a concentration of 10^{-6} nm^{-3} and a distance of 50 nm below the surface. Hence, in this simulation the dilute approximation breaks down well before bubble bursting events take place.

3.2.2. Influence of temperature. Figures 13 and 14 show the strong effect of temperature on the model predictions. In the case of helium implantation at 500 K, the lower diffusivity of helium interstitials produces a faster rate of self-trapping and clustering which occurs closer to the surface: the model

(a) Concentration of large HeV clusters ($\geq He_{250}V_{50}$)

(b) Average radius



(c) Volume fraction

Figure 13. Concentration of large HeV clusters (a), average radius (b) and volume fraction (c) under the tungsten surface after $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium implantation at 500 K up to a fluence of $2 \times 10^{22} \text{ m}^{-2}$.

predicts high concentrations ($\geq 10^{-4} \text{ nm}^{-3}$) of large clusters (radius $\geq 10 \text{ nm}$) close to the surface for very low fluences ($\approx 10^{22} \text{ m}^{-2}$). Based on MD simulations, we would then expect bubble bursting phenomenon to occur before the dilute limit

breaks down and bubble coalescence takes place. We also do not observe a plug of helium bubbles propagating towards the surface, as was observed at higher temperature since this bubble layer has already formed close to the implantation

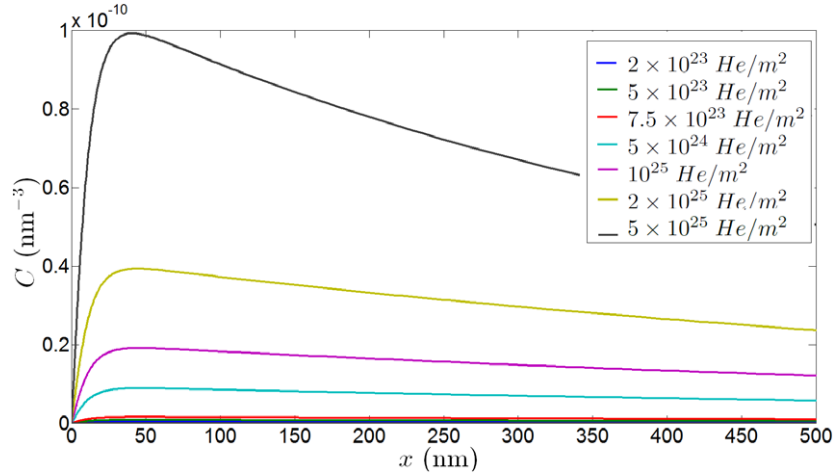


Figure 14. Concentration of large HeV clusters ($\geq \text{He}_{250}\text{V}_{50}$) under the tungsten surface after $10^{22} \text{ m}^{-2} \text{ s}^{-1}$ 100 eV helium implantation at 1500 K up to a fluence of $5 \times 10^{25} \text{ m}^{-2}$.

layer. In the case of irradiation at 1500 K, the enhanced mobility drastically lowers the amount of clustering and self-trapping. Therefore, no significant amount of helium bubbles are predicted to form below the tungsten surface even at high fluences ($\geq 5 \times 10^{25} \text{ m}^{-2}$).

These results predict that lower temperature and higher flux produce higher concentrations of helium bubbles with larger radius, which are predicted to form closer to the surface. Therefore, we would expect significant surface morphology changes to occur for the lowest temperatures. However, experimental evidence shows the opposite effect; Miyamoto *et al* [20] observe a large concentration of small nanometre-sized helium bubbles forming under the tungsten surface at 573 K under combined helium and deuterium implantation. The bubble concentration is above the percolation threshold and thus, the bubbles are interlinked. However, no significant surface morphology change is observed. On the other hand, Baldwin *et al* [19] and Kajita *et al* [1] observe very strong surface morphology changes (e.g. ‘nanofuzz’ formation) for temperatures as high as 1400 K. A possible explanation for this behaviour is that at low temperatures ($T \sim 500 \text{ K}$), bubble bursting is impeded by a rapid coalescence of helium bubbles below the surface which relieves the pressure, while at high temperatures ($T > 1000 \text{ K}$), bubble bursting events are not impeded due to the lower concentration of helium bubbles and hence the surface morphology is strongly affected. However, this discrepancy requires further research and analysis.

3.2.3. Comparison with an analytical model. These results are again in very good agreement with an analytical model proposed by Krasheninnikov [10], which predicts the same behaviour with respect to a front of large gas bubbles propagating towards the surface followed by a region depleted in mobile helium-interstitial concentration at deeper depths than this front. The analytical model makes simplifying assumptions such as neglecting dissociation of helium–vacancy clusters and a steady-state density of single helium atoms in the implantation layer based on dimensional considerations, but the same reaction–diffusion description of microstructure evolution is used. The qualitative picture of the

analytical model is shown in figure 15. We observe very similar results as those predicted by our model shown in figure 10: the rapidly established steady-state concentrations of single helium interstitials is depleted by the build-up of a plug of large helium–vacancy clusters. The main discrepancy observed is the smaller depletion of the single helium concentration at the surface boundary in our model compared to the analytical model prediction. This can be explained by the simplifying assumption made by the analytical model about the steady-state concentration of single helium interstitials in the boundary layer.

3.2.4. Model limitations. Several factors are not yet considered within this model and will be the object of further study: The high volume fraction makes the results of the cluster dynamics model less reliable for high fluences in the low temperature ($T \sim 500 \text{ K}$) range. Therefore, the growth of moderately large gas bubbles in a high density environment should be further investigated using multiscale modelling techniques. The surface proximity and the possibility of drift diffusion forces acting on helium and small clusters have also not been considered in this work.

In the high temperature range ($T \sim 1500 \text{ K}$), the model predicts a much lower helium bubble density than what is observed in experiments. A possible reason is that the model neglects impurities in the tungsten materials, which can act as trapping sites for helium-interstitial atoms and possibly enhance the nucleation of small He_xV_y clusters. Further investigation of the properties of impurity–helium interactions is therefore recommended, as well as additional research to validate the helium–vacancy thermodynamic and kinetic parameters used in this work.

4. Conclusion

We have described a cluster dynamics model to simulate low-energy helium implantation in tungsten under fusion relevant conditions at high fluxes and temperatures. The key elements of the model are the high mobility of helium-interstitial atoms in the tungsten matrix, the self-trapping of helium atoms to

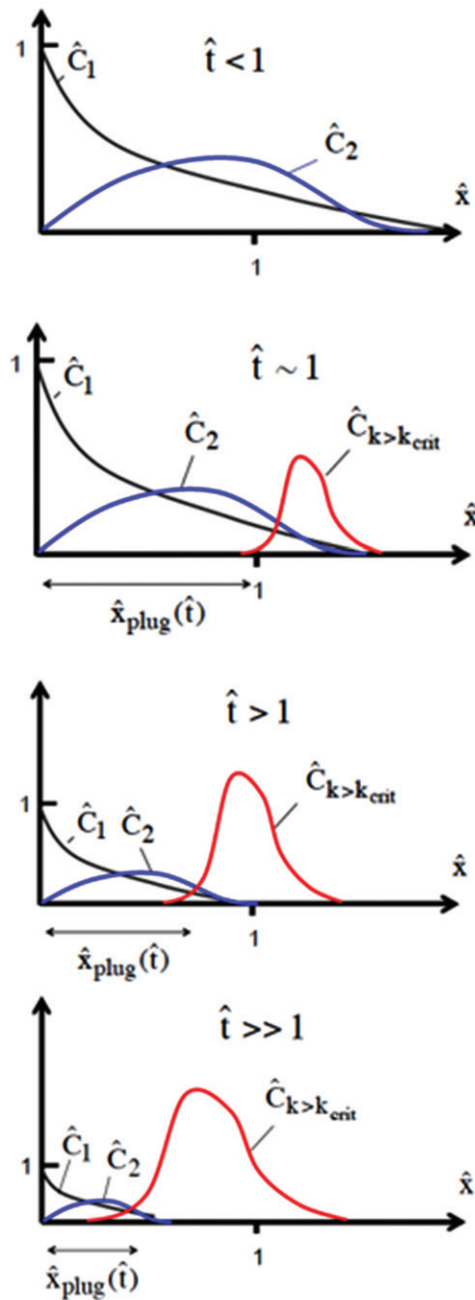


Figure 15. Qualitative picture of the evolution of the concentration of the helium bubbles. $\hat{C}$, $\hat{t}$ and $\hat{x}$ are dimensionless quantities depending on the flux and temperature. Reproduced with permission from [10]. Copyright 2014 IAEA.

form mobile helium clusters, and the trap mutation process to grow larger helium–vacancy clusters. The model is in excellent agreement with an analytical model developed by Krashenninnikov and presented in [10]. We predict that self-trapping of helium atoms does not occur in tungsten tendrils at temperatures where tungsten nanofuzz is observed to form, and hence the observed helium gas bubbles should be formed directly under the surface before surface morphology changes occur. For implantation of helium into a perfect tungsten surface with an infinite half-slab geometry, the model identifies different regimes depending on the temperature: for a given

fluence, large concentrations of helium gas bubbles are formed close to the surface at low temperatures ($T \approx 500$ K), while lower concentrations of smaller gas bubbles are formed deeper into the material at intermediate temperatures ($T \approx 1000$ K). The model ceases to be valid at a temperature dependent fluence due to either bubble bursting phenomena expected to take place near the surface (but not explicitly modelled here) or due to a high volume fraction of helium species leading to bubble coalescence. At higher temperatures (1500 K), the model predicts a much smaller concentration of large helium–vacancy clusters, in apparent disagreement with experimental observations, which might be caused by not taking into account the effect of the various impurities in the tungsten material. These processes will be the object of further investigation in future work.

Acknowledgments

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References

- [1] Kajita S., Sakaguchi W., Ohno N., Yoshida N. and Saeki T. 2009 Formation process of tungsten nanostructure by the exposure to helium plasma under fusion relevant plasma conditions *Nucl. Fusion* **49** 095005
- [2] Gilliam S.B., Gidcumb S.M., Parikh N.R., Forsythe D.G., Patnaik B.K., Hunn J.D., Snead L.L. and Lamaze G.P. 2005 Retention and surface blistering of helium irradiated tungsten as a first wall material *J. Nucl. Mater.* **347** 289–97
- [3] Hino T., Koyama K., Yamauchi Y. and Hirohata Y. 1998 Hydrogen retention properties of polycrystalline tungsten and helium irradiated tungsten *Fusion Eng. Des.* **39–40** 227–33
- [4] Xu D., Wirth B.D., Li M. and Kirk M.A. 2012 Combining *in situ* transmission electron microscopy irradiation experiments with cluster dynamics modeling to study nanoscale defect agglomeration in structural metals *Acta Mater.* **60** 4286–302
- [5] Ortiz C.J. and Caturia M.J. 2007 Simulation of defect evolution in irradiated materials: role of intracascade clustering, correlated recombination *Phys. Rev. B* **75** 184101
- [6] Meslin E., Barbu A., Boulanger L., Radiguet B., Pareige P., Arakawa K. and Fu C.C. 2008 Cluster-dynamics modelling of defects in α -iron under cascade damage conditions *J. Nucl. Mater.* **382** 190–6
- [7] Li Y.G., Zhou W.H., Huang L.F., Zeng Z. and Ju X. 2012 Cluster dynamics modeling of accumulation and diffusion of helium in neutron irradiated tungsten *J. Nucl. Mater.* **431** 26–32
- [8] Watanabe Y., Iwakiri H., Yoshida N., Morishita K. and Kohyama A. 2007 Formation of interstitial loops in tungsten under helium ion irradiation: rate theory modeling and experiment *Nucl. Instrum. Methods Phys. Res. B* **255** 32–6
- [9] Faney T. and Wirth B.D. 2014 Spatially dependent cluster dynamics modeling of microstructure evolution in low

- energy helium irradiated tungsten *Modelling Simul. Mater. Sci. Eng.* **22** 065010
- [10] Krashennnikov S.I., Faney T. and Wirth B.D. 2014 On helium cluster dynamics in tungsten plasma facing components of fusion devices *Nucl. Fusion* **54** 073019
- [11] Sefta F., Hammond K.D., Juslin N. and Wirth B.D. 2013 Tungsten surface evolution by helium bubble nucleation, growth and rupture *Nucl. Fusion* **53** 073015
- [12] Ventelon L., Wirth B. and Domain C. 2006 Helium self-interstitial atom interaction in α -iron *J. Nucl. Mater.* **351** 119–32
- [13] Kajita S., Yoshida N., Yoshihara R., Ohno N. and Yamagiwa M. 2011 {TEM} observation of the growth process of helium nanobubbles on tungsten: nanostructure formation mechanism *J. Nucl. Mater.* **418** 152–8
- [14] Ziegler J.F., Ziegler M.D. and Biersack J.P. 2010 SRIM—the stopping and range of ions in matter *Nucl. Instrum. Methods Phys. Res. B* **268** 1818–23
- [15] Waite T.R. 1957 Theoretical treatment of the kinetics of diffusion-limited reactions *Phys. Rev.* **107** 463–70
- [16] Lasa A., Henriksson K.O.E. and Nordlund K. 2013 {MD} simulations of onset of tungsten fuzz formation under helium irradiation *Nucl. Instrum. Methods Phys. Res. B* **303** 156–61
- [17] Juslin N. and Wirth B.D., in preparation
- [18] Faney T. and Wirth B.D. 2014 Efficient implementation of cluster dynamics modeling for large systems of reaction–diffusion equations *J. Comput. Phys.* submitted
- [19] Baldwin M.J. and Doerner R.P. 2008 Helium induced nanoscopic morphology on tungsten under fusion relevant plasma conditions *Nucl. Fusion* **48** 035001
- [20] Miyamoto M., Nishijima D., Baldwin M.J., Doerner R.P., Ueda Y., Yasunaga K., Yoshida N. and Ono K. 2011 Microscopic damage of tungsten exposed to deuterium–helium mixture plasma in {PISCES} and its impacts on retention property *J. Nucl. Mater.* **415** (Suppl.) S657–60
- [21] Wirth B.D. 2013 Manuscript in submission *11th Int. Symp. on Fusion Nuclear Technology (Barcelona, Spain, 16–20 September 2013)* www.isfnt-11.org/welcome.html